

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE COURSE [PHASE-III]

TEST # 08

DATE : 02-12-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 08

Topics

DATE : 02 - 12 - 2018

PRE-MEDICAL NURTURE COURSE [PHASE-III] : [NEET]

PHYSICS : Centre of Mass, Collision, Rotational Motion, SHM

CHEMISTRY : Chemical Bonding, Thermodynamics, Chemical & Ionic Equilibrium

BIOLOGY : Enzymes, Plant Anatomy, Plant Morphology, Frog, Digestive System,
Respiratory System

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

- | | |
|---|--|
| <p>1. A simple pendulum performs simple harmonic motion about $X = 0$ with an amplitude A and time period T. The speed of the pendulum at $X = \frac{A}{2}$ will be</p> <p>(1) $\frac{\pi A \sqrt{3}}{T}$ (2) $\frac{\pi A}{T}$</p> <p>(3) $\frac{\pi A \sqrt{3}}{2T}$ (4) $\frac{3\pi^2 A}{T}$</p> <p>2. A body of mass 5 gm is executing S.H.M. about a point with amplitude 10 cm. Its maximum velocity is 100 cm/sec. Its velocity will be 50 cm/sec at a distance</p> <p>(1) 5 (2) $5\sqrt{2}$</p> <p>(3) $5\sqrt{3}$ (4) $10\sqrt{2}$</p> <p>3. A particle executes S.H.M. with a period of 6 second and amplitude of 3 cm. Its maximum speed in cm/sec is</p> <p>(1) $\pi/2$ (2) π</p> <p>(3) 2π (4) 3π</p> <p>4. A body is executing S.H.M. When its displacement from the mean position is 4 cm and 5 cm, the corresponding velocity of the body is 10 cm/sec and 8 cm/sec. Then the time period of the body is</p> <p>(1) 2π sec (2) $\pi/2$ sec</p> <p>(3) π sec (4) $3\pi / 2$ sec</p> | <p>5. If a simple pendulum oscillates with an amplitude of 50 mm and time period of 2 sec, then its maximum velocity is</p> <p>(1) 0.10 m / s</p> <p>(2) 0.15 m / s</p> <p>(3) 0.8 m / s</p> <p>(4) 0.26 m / s</p> <p>6. A particle executes simple harmonic motion with an amplitude of 4 cm. At the mean position the velocity of the particle is 10 cm/s. The distance of the particle from the mean position when its speed becomes 5 cm/s is</p> <p>(1) $\sqrt{3}$ cm (2) $\sqrt{5}$ cm</p> <p>(3) $2(\sqrt{3})$ cm (4) $2(\sqrt{5})$ cm</p> <p>7. The amplitude of a particle executing SHM is 4 cm. At the mean position the speed of the particle is 16 cm/sec. The distance of the particle from the mean position at which the speed of the particle becomes $8\sqrt{3}$ cm/s, will be</p> <p>(1) $2\sqrt{3}$ cm (2) $\sqrt{3}$ cm</p> <p>(3) 1 cm (4) 2 cm</p> <p>8. The displacement equation of a particle is $x = 3\sin 2t + 4\cos 2t$. The amplitude and maximum velocity will be respectively</p> <p>(1) 5, 10 (2) 3, 2</p> <p>(3) 4, 2 (4) 3, 4</p> |
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SPACE FOR ROUGH WORK

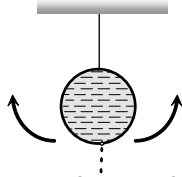
9. The instantaneous displacement of a simple pendulum oscillator is given by $x = A \cos\left(\omega t + \frac{\pi}{4}\right)$. Its speed will be maximum at time :-
 (1) $\frac{\pi}{4\omega}$ (2) $\frac{\pi}{2\omega}$ (3) $\frac{\pi}{\omega}$ (4) $\frac{2\pi}{\omega}$
10. A small body of mass 0.10 kg is executing S.H.M. of amplitude 1.0 m and period 0.20 sec. The maximum force acting on it is
 (1) 98.596 N (2) 985.96 N
 (3) 100.2 N (4) 76.23 N
11. A body executing simple harmonic motion has a maximum acceleration equal to 24 metres/sec² and maximum velocity equal to 16 metres/sec. The amplitude of the simple harmonic motion is
 (1) $\frac{32}{3}$ metres (2) $\frac{3}{32}$ metres
 (3) $\frac{1024}{9}$ metres (4) $\frac{64}{9}$ metres
12. A particle of mass 10 grams is executing simple harmonic motion with an amplitude of 0.5 m and periodic time of $(\pi/5)$ seconds. The maximum value of the force acting on the particle is
 (1) 25 N (2) 5 N
 (3) 2.5 N (4) 0.5 N
13. A body is vibrating in simple harmonic motion with an amplitude of 0.06 m and frequency of 15 Hz. The velocity and acceleration of body is
 (1) 5.65 m/s and 5.32×10^2 m/s²
 (2) 6.82 m/s and 7.62×10^2 m/s²
 (3) 8.91 m/s and 8.21×10^2 m/s²
 (4) 9.82 m/s and 9.03×10^2 m/s²
14. A particle executes linear simple harmonic motion with an amplitude of 2 cm. When the particle is at 1 cm from the mean position the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is
 (1) $\frac{1}{2\pi\sqrt{3}}$ (2) $2\pi\sqrt{3}$
 (3) $\frac{2\pi}{\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2\pi}$
15. A particle is vibrating in a simple harmonic motion with an amplitude of 4 cm. At what displacement from the equilibrium position, is its energy half potential and half kinetic
 (1) 1 cm (2) $\sqrt{2}$ cm
 (3) 3 cm (4) $2\sqrt{2}$ cm
16. The kinetic energy and potential energy of a particle executing simple harmonic motion will be equal, when displacement (amplitude = a) is
 (1) $\frac{a}{2}$ (2) $a\sqrt{2}$
 (3) $\frac{a}{\sqrt{2}}$ (4) $\frac{a\sqrt{2}}{3}$

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|---|---|
| <p>17. The total energy of the body executing S.H.M. is E. Then the kinetic energy when the displacement is half of the amplitude, is</p> <p>(1) $\frac{E}{2}$ (2) $\frac{E}{4}$</p> <p>(3) $\frac{3E}{4}$ (4) $\frac{\sqrt{3}}{4}E$</p> <p>18. The potential energy of a particle executing S.H.M. is 2.5 J, when its displacement is half of amplitude. The total energy of the particle be</p> <p>(1) 18 J (2) 10 J</p> <p>(3) 12 J (4) 2.5 J</p> <p>19. The angular velocity and the amplitude of a simple pendulum is ω and a respectively. At a displacement X from the mean position if its kinetic energy is T and potential energy is V, then the ratio of T to V is</p> <p>(1) $X^2\omega^2/(a^2 - X^2\omega^2)$ (2) $X^2/(a^2 - X^2)$</p> <p>(3) $(a^2 - X^2\omega^2)$ (4) $(a^2 - X^2)/X^2$</p> <p>20. The total energy of a particle executing S.H.M. is 80 J. What is the potential energy when the particle is at a distance of $3/4$ of amplitude from the mean position</p> <p>(1) 60 J (2) 10 J</p> <p>(3) 40 J (4) 45 J</p> <p>21. A body executes simple harmonic motion. The potential energy (P.E.), the kinetic energy (K.E.) and total energy (T.E.) are measured as a function of displacement x. Which of the following statements is true</p> | <p>(1) P.E. is maximum when $x = 0$</p> <p>(2) K.E. is maximum when $x = 0$</p> <p>(3) T.E. is zero when $x = 0$</p> <p>(4) K.E. is maximum when x is maximum</p> <p>22. The total energy of a particle, executing simple harmonic motion is</p> <p>(1) $\propto x$</p> <p>(2) $\propto x^2$</p> <p>(3) Independent of x</p> <p>(4) $\propto x^{1/2}$</p> <p>23. The kinetic energy of a particle executing S.H.M. is 16 J when it is at its mean position. If the mass of the particle is 0.32 kg, then what is the maximum velocity of the particle</p> <p>(1) 5 m/s (2) 15 m/s</p> <p>(3) 10 m/s (4) 20 m/s</p> <p>24. The equation of motion of a particle is $\frac{d^2y}{dt^2} + Ky = 0$, where K is positive constant. The time period of the motion is given by</p> <p>(1) $\frac{2\pi}{K}$ (2) $2\pi K$</p> <p>(3) $\frac{2\pi}{\sqrt{K}}$ (4) $2\pi\sqrt{K}$</p> <p>25. The acceleration of a particle performing S.H.M. is 12 cm/sec^2 at a distance of 3 cm from the mean position. Its time period is :-</p> <p>(1) 0.5 sec (2) 1.0 sec</p> <p>(3) 2.0 sec (4) 3.14 sec</p> |
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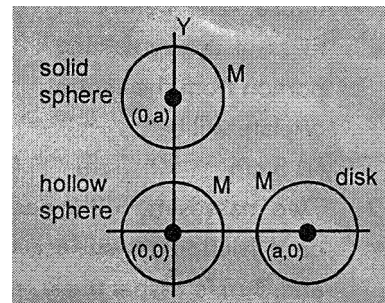
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26. A simple pendulum is made of a body which is a hollow sphere containing mercury suspended by means of a wire. If a little mercury is drained off, the period of pendulum will



- (1) Remains unchanged
 - (2) First increase then decrease to initial value
 - (3) Decrease
 - (4) Become erratic
27. A pendulum suspended from the ceiling of a train has a period T , when the train is at rest. When the train is accelerating with a uniform acceleration a , the period of oscillation will
- (1) Increase
 - (2) Decrease
 - (3) Remain unaffected
 - (4) Become infinite
28. A simple pendulum is set up in a trolley which moves to the right with an acceleration a on a horizontal plane. Then the thread of the pendulum in the mean position makes an angle θ with the vertical :-
- (1) $\tan^{-1} \frac{a}{g}$ in the forward direction
 - (2) $\tan^{-1} \frac{a}{g}$ in the backward direction
 - (3) $\tan^{-1} \frac{g}{a}$ in the backward direction
 - (4) $\tan^{-1} \frac{g}{a}$ in the forward direction

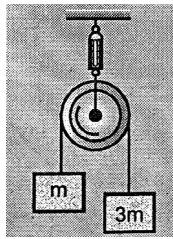
29. The time period of a second's pendulum is 2 sec. The spherical bob which is empty from inside has a mass of 50 gm. This is now replaced by another solid bob of same radius but having different mass of 100 gm. The new time period will be
- (1) 4 sec
 - (2) 1 sec
 - (3) 2 sec
 - (4) 8 sec
30. The coordinate of the centre of mass of a system as shown in figure:-



- (1) $\left(\frac{a}{3}, 0\right)$
 - (2) $\left(\frac{a}{2}, \frac{a}{2}\right)$
 - (3) $\left(\frac{a}{3}, \frac{a}{3}\right)$
 - (4) $\left(0, \frac{a}{3}\right)$
31. The centre of mass of a system of three particles of masses $1g$, $2g$ and $3g$ is taken as the origin of a coordinate system. The position vector of a fourth particle of mass $4g$ such that the centre of mass of the four particle system lies at the point $(1, 2, 3)$ is $\alpha(\hat{i} + 2\hat{j} + 3\hat{k})$, where α is a constant. The value of α is :-
- (1) $\frac{10}{3}$
 - (2) $\frac{5}{2}$
 - (3) $\frac{1}{2}$
 - (4) $\frac{2}{5}$

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32. If the system is released, then the acceleration of the centre of mass of the system :-



- (1) $\frac{g}{4}$ (2) $\frac{g}{2}$
(3) g (4) $2g$
33. A heavy nucleus at rest breaks into two fragments which fly off with velocities 8 : 1. The ratio of radii of the fragments is :-
(1) 1 : 2 (2) 1 : 4
(3) 4 : 1 (4) 2 : 1
34. A bullet of mass m is fired from a gun of mass M . The recoiling gun compresses a spring of force constant k by a distance d . Then the velocity of the bullet is :-
(1) $kd\sqrt{M/m}$
(2) $\frac{d}{M}\sqrt{km}$
(3) $\frac{d}{m}\sqrt{kM}$
(4) $\frac{kM}{m}\sqrt{d}$
35. A sphere of mass m moving with a constant velocity collides with another stationary sphere of same mass. The ratio of velocities of two spheres after collision will be, if the co-efficient of restitution is e :-

(1) $\frac{1-e}{1+e}$ (2) $\frac{e-1}{e+1}$

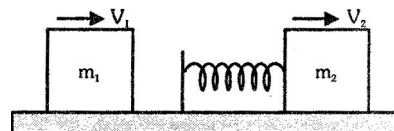
(3) $\frac{1+e}{1-e}$ (4) $\frac{e+1}{e-1}$

36. A particle falls from a height 'h' upon a fixed horizontal plane and rebounds. If V is the coefficient of restitution the total distance travelled before rebounding has stopped is :-

(1) $h\left(\frac{1+e^2}{1-e^2}\right)$ (2) $h\left(\frac{1-e^2}{1+e^2}\right)$

(3) $\frac{h}{2}\left(\frac{1-e^2}{1+e^2}\right)$ (4) $\frac{h}{2}\left(\frac{1+e^2}{1-e^2}\right)$

37. Two masses $m_1 = 2\text{kg}$ and $m_2 = 5\text{kg}$ are moving on a frictionless surface with velocities 10 m/s and 3 m/s respectively. m_2 is ahead of m_1 . An ideal spring of spring constant $k = 1120\text{ N/m}$ is attached on the back side of m_2 . The maximum compression of the spring will be :-



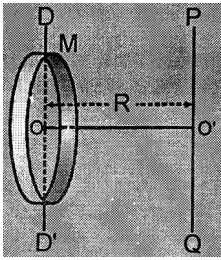
(1) 0.51 m (2) 0.062 m
(3) 0.25 m (4) 0.72 m

38. A solid sphere and a hollow sphere of the same mass have the same M.I. about their respective diameters. The ratio of their radii will be :-

(1) 1 : 2 (2) $\sqrt{3} : \sqrt{5}$
(3) $\sqrt{5} : \sqrt{3}$ (4) 4 : 5

SPACE FOR ROUGH WORK

39. The moment of inertia of a ring of mass M and radius R about PQ axis will be :-



- (1) MR^2 (2) $\frac{MR^2}{2}$
(3) $\frac{3}{2}MR^2$ (4) $2MR^2$
40. A constant torque acting on a uniform circular wheel changes its angular momentum from A_0 to $4A_0$ in 4 seconds. The magnitude of this torque is :-
- (1) $\frac{3A_0}{4}$ (2) A_0
(3) $4A_0$ (4) $12A_0$
41. A round disc of moment of inertia I_2 about its axis perpendicular to its plane and passing through its centre is placed over another disc of moment of inertia I_1 rotating with an angular velocity ω about the same axis. The final angular velocity of the combination of discs is :-
- (1) ω (2) $\frac{I_1\omega}{I_1 + I_2}$
(3) $\frac{(I_1 + I_2)\omega}{I_1}$ (4) $\frac{I_2\omega}{I_1 + I_2}$

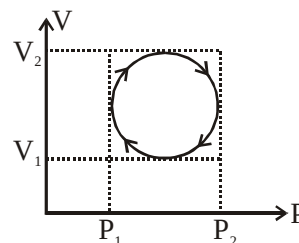
42. A small steel sphere of mass m is tied to a string of length r and is whirled in a horizontal circle with a uniform angular velocity 2ω . The string is suddenly pulled, so that radius of the circle is halved. The new angular velocity will be
- (1) 2ω (2) 4ω
(3) 6ω (4) 8ω
43. A ring is rolling without slipping. Its energy of translation is E . Its total kinetic energy will be :-
- (1) E (2) $2E$
(3) $3E$ (4) $4E$
44. If rotational kinetic energy is 50% of total kinetic energy then the body will be :-
- (1) ring
(2) cylinder
(3) hollow sphere
(4) solid sphere
45. Two rotating bodies A and B of masses m and $2m$ with moments of inertia I_A and I_B ($I_B > I_A$) have equal kinetic energy of rotation. If L_A and L_B be their angular momenta respectively, then :-
- (1) $L_B > L_A$
(2) $L_A > L_B$
(3) $L_A = \frac{L_B}{2}$
(4) $L_A = 2L_B$

SPACE FOR ROUGH WORK

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| <p>46. For the reaction, $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, the equilibrium constant K_p changes with:</p> <p>(1) total pressure
 (2) catalyst
 (3) the amounts of H_2 and I_2 present
 (4) temperature</p> <p>47. In the reaction, $A_2(g) + 4B_2(g) \rightleftharpoons 2AB_4(g)$, $\Delta H < 0$, the formation of $AB_4(g)$ will be favoured at:</p> <p>(1) low T and high P
 (2) high T and low P
 (3) low T and low P
 (4) high T and high P</p> <p>48. For which of the following reactions, $K_p = K_c$?</p> <p>(1) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$
 (2) $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$
 (3) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
 (4) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$</p> <p>49. Densities of diamond and graphite are 3.5 and 2.3 g/mL respectively. Increase of pressure on the equilibrium $C(\text{diamond}) \rightleftharpoons C(\text{graphite})$</p> <p>(1) favours backward reaction
 (2) favours forward reaction
 (3) has no effect
 (4) increase the reaction rate</p> <p>50. Two solid compounds X(s) and Y(s) undergo dissociation as follows :</p> <p>$X(s) \rightleftharpoons A(g) + 2B(g); K_{p_1} = 9 \times 10^{-3} \text{ bar}^3$</p> <p>$Y(s) \rightleftharpoons 2B(g) + C(g); K_{p_2} = 4.5 \times 10^{-3} \text{ bar}^3$</p> <p>The total pressure of gaseous mixture at equilibrium will be :</p> | <p>(1) 4.5 bar (2) 0.45 bar
 (3) 0.6 bar (4) 1.2 bar</p> <p>51. When equal volumes of the following solutions are mixed, precipitation of $AgCl$ ($K_{sp} = 1.8 \times 10^{-10}$) will occur only with:</p> <p>(1) $10^{-4}M(Ag^+)$ and $10^{-4}M(Cl^-)$
 (2) $10^{-5}M(Ag^+)$ and $10^{-5}M(Cl^-)$
 (3) $10^{-6}M(Ag^+)$ and $10^{-6}M(Cl^-)$
 (4) $10^{-10}M(Ag^+)$ and $10^{-10}M(Cl^-)$</p> <p>52. Which of the following solutions will have pH close to 1.0?</p> <p>(1) 100 mL of (M/10) HCl + 100 mL of (M/10) NaOH
 (2) 55 mL of (M/10) HCl + 45 mL of (M/10) NaOH
 (3) 10 mL of (M/10) HCl + 90 mL of (M/10) NaOH
 (4) 75 mL of (M/5) HCl + 25 mL of (M/5) NaOH</p> <p>53. The solubility of A_2X_3 is $y \text{ mol dm}^{-3}$. Its solubility product is :-</p> <p>(1) $6y^4$ (2) $64y^4$
 (3) $36y^5$ (4) $108y^5$</p> <p>54. How many moles of $HCOONa$ must be added to 1 L of 0.1 M $HCOOH$ to prepare a buffer solution with a pH of 3.4? (Given : K_a for $HCOOH = 2 \times 10^{-4}$)</p> <p>(1) 0.01 (2) 0.05
 (3) 0.1 (4) 0.2</p> |
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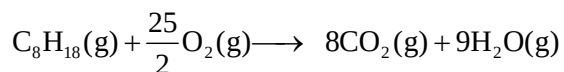
55. 0.1 M HCl and 0.1 M H_2SO_4 each of volume 2 mL are mixed and the volume is made upto 6 mL by adding 2 mL of 0.01 N NaCl solution. The pH of the resulting mixture is:
 (1) 1.17 (2) 1.0
 (3) 0.3 (4) $\log 2 - \log 3$
56. For the reaction,
 $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g); \Delta H^\circ = -ve$
 the number of moles of H_2 at equilibrium will increase when :-
 (1) volume of vessel increased
 (2) volume of vessel is decreased
 (3) Ne gas is added at constant volume
 (4) NH_3 is removed
57. Conjugate base of $H_2PO_4^-$ is :-
 (1) H_3PO_4 (2) $H_2PO_3^-$
 (3) HPO_4^{2-} (4) PO_4^{3-}
58. The number of OH^- ion in 1 mL of solution having pH = 4 is :-
 (1) 10^{-4}
 (2) 10^{-10}
 (3) 6.02×10^{10}
 (4) 6.02×10^{13}
59. The pH of 10^{-10} M H_2SO_4 will be almost :-
 (1) 4 (2) 7
 (3) 6 (4) 0
60. Select the anion which is the strongest bronsted base :-
 (1) ClO_4^- (2) ClO_3^-
 (3) ClO_2^- (4) ClO^-
61. In which of the following pairs, both properties are intensive?
 (1) Pressure, mass
 (2) Density, volume
 (3) Temperature, density
 (4) Pressure, volume
62. The enthalpy of neutralization of a strong acid by a strong base is $-57.32 \text{ kJ mol}^{-1}$. The enthalpy of formation of water is $-285.84 \text{ kJ mol}^{-1}$. The enthalpy of formation of hydroxyl ion is :-
 (1) $+228.52 \text{ kJ mol}^{-1}$
 (2) $-114.26 \text{ kJ mol}^{-1}$
 (3) $-228.52 \text{ kJ mol}^{-1}$
 (4) $+114.2 \text{ kJ mol}^{-1}$
63. In the cyclic process shown in P - V diagram, the magnitude of the work done is :-



- (1) $\pi \left(\frac{P_2 - P_1}{2} \right)^2$
 (2) $\pi \left(\frac{V_2 - V_1}{2} \right)^2$
 (3) $\frac{\pi}{4} (P_2 - P_1)(V_2 - V_1)$
 (4) $\pi (V_2 - V_1)^2$

SPACE FOR ROUGH WORK

64. Combustion of octane takes place in an automobile engine. The homogeneous equation of combustion is :



The signs of ΔH , ΔS and ΔG for the reaction will be :-

- (1) +ve, -ve, +ve
(2) -ve, +ve, -ve
(3) -ve, +ve, +ve
(4) +ve, +ve, -ve
65. A gas expands adiabatically at constant pressure such that :

$$T \propto \frac{1}{\sqrt{V}}$$

The value of γ , i.e., (C_p/C_v) of the gas will be :-

- (1) 1.30 (2) 1.50
(3) 1.70 (4) 2
66. The correct order of decreasing polarisability of ion is :-
- (1) $\text{Cl}^- > \text{Br}^- > \text{I}^- > \text{F}^-$
(2) $\text{F}^- > \text{I}^- > \text{Br}^- > \text{Cl}^-$
(3) $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$
(4) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$

67. Which ion has a higher polarising power ?
- (1) Mg^{2+} (2) Al^{3+}
(3) Ca^{2+} (4) Na^+

68. The correct order for bond angles is :-

- (1) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$
(2) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_2$
(3) $\text{NO}_2 > \text{NO}_2^- > \text{NO}_2^+$
(4) $\text{NO}_2^- > \text{NO}_2 > \text{NO}_2^+$

69. The shapes of PCl_4^+ , PCl_4^- and AsCl_5 are respectively :-

- (1) square planar, tetrahedral, see-saw
(2) tetrahedral, see-saw, trigonal bipyramidal
(3) tetrahedral, square planar and pentagonal bipyramidal
(4) trigonal bipyramidal, tetrahedral and square pyramidal

70. The diamagnetic molecules are :-

- (1) B_2 , C_2 , N_2 (2) O_2 , N_2 , F_2
(3) C_2 , N_2 , F_2 (4) B_2 , O_2^{2-} , N_2

71. The correct bond order for CO and CO^+ are respectively :-

- (1) 3, 5/2 (2) 3, 2
(3) 3, 7/2 (4) 5/2, 3

72. In the series ethane, ethylene and acetylene, the C-H bond energy is :-

- (1) the same in all the three compounds
(2) greatest in ethane
(3) greatest in ethylene
(4) greatest in acetylene

73. The bond angle is minimum in :-

- (1) H_2Te (2) H_2Se
(3) H_2O (4) H_2S

SPACE FOR ROUGH WORK

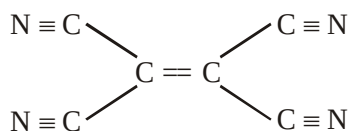
74. In an octahedral structure, the pair of d-orbitals involved in sp^3d^2 -hybridisation is :-

- (1) d_{x^2}, d_{xz}
- (2) d_{xy}, d_{yz}
- (3) $d_{x^2-y^2}, d_{z^2}$
- (4) $d_{xz}, d_{x^2-y^2}$

75. The correct order of increasing C – O bond length of CO , CO_3^{2-} , CO_2 is :-

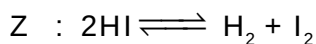
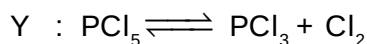
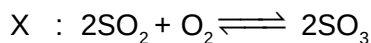
- (1) $CO_3^{2-} < CO_2 < CO$
- (2) $CO_2 < CO_3^{2-} < CO$
- (3) $CO < CO_3^{2-} < CO_2$
- (4) $CO < CO_2 < CO_3^{2-}$

76. How many σ and π bonds are there in the molecule of tetracyanoethylene?



- (1) Nine σ and nine π
- (2) Five σ and nine π
- (3) Nine σ and seven π
- (4) Five σ and eight π

77. For the following gaseous equilibria X, Y and Z at 300 K



ratio of K_p and K_c in the increasing order is

- (1) $X = Y = Z$
- (2) $X < Y < Z$
- (3) $X < Z < Y$
- (4) $Z < Y < X$

78. For a very small extent of dissociation of PCl_5 into PCl_3 and Cl_2 is a gaseous phase reaction then degree of dissociation x

- (1) $x \propto p$
- (2) $x \propto \frac{1}{p}$

$$(3) x \propto \sqrt{p}$$

$$(4) x \propto \sqrt{\frac{1}{p}}$$

79. 60 g CH_3COOH and 46 g CH_3CH_2OH react in 5L flask to form

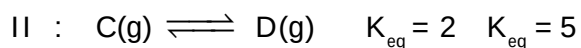
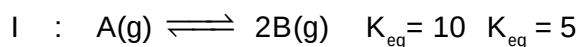
44 g $CH_3COOC_2H_5$ at equilibrium. On taking 120 g CH_3COOH and 46 g CH_3CH_2OH , $CH_3COOC_2H_5$ formed at equilibrium is

- (1) 44 g
- (2) 20.33 g
- (3) 22 g
- (4) 58.66 g

SPACE FOR ROUGH WORK

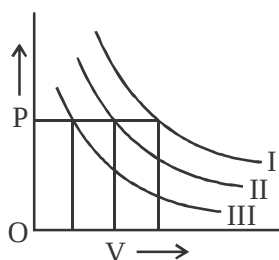
80. Consider the following equilibria at 300 K and 400 K with their equilibrium constants

300 K 400 K

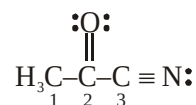


Thus,

- (1) I is endothermic, II is exothermic
 - (2) I is exothermic, II is endothermic
 - (3) I and II both are endothermic
 - (4) I and II both are exothermic
81. 50.0 mL of 0.3 M HCl is mixed with 50 mL of 0.4 M NH_3 solution. If pK_a of NH_4^+ is 9.26, pH of the mixture is
- (1) 5.22
 - (2) 1.30
 - (3) 8.78
 - (4) 12.70
82. K_{sp} of $\text{Al}(\text{OH})_3$ is 1.0×10^{-15} . pH of the saturated solution is about
- (1) 5.0
 - (2) 9.0
 - (3) 4.1
 - (4) 10.4
83. At 4°C , $K_w = 1 \times 10^{-16}$. A solution with $\text{pH} = 7.5$ at 4°C will
- (1) turn blue litmus red
 - (2) turn red litmus blue
 - (3) turn turmeric paper brown
 - (4) be neutral to litmus
84. I, II, III are three isotherms respectively at T_1 , T_2 and T_3 . Temperature will be in order



85. Which pair is not correct order of lattice energy?
- (1) $T_1 = T_2 = T_3$ (2) $T_1 < T_2 < T_3$
(3) $T_1 > T_2 > T_3$ (4) $T_1 > T_2 = T_3$
86. What hybrid orbitals are employed by carbon atoms 1, 2 and 3, respectively as labelled in the compound shown?



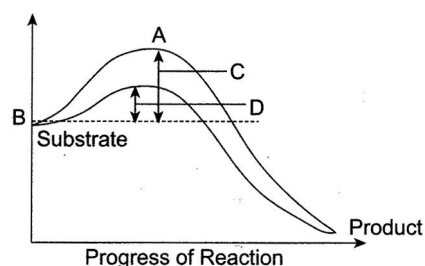
- (1) sp^3 , sp , sp (2) sp^3 , sp^2 , sp
(3) sp^3 , sp , sp^2 (4) sp^3 , sp^2 , sp^2
87. Which pair of substances will have the most similar geometry?
- (1) SO_3 , SO_3^{2-} (2) SO_3 , SO_4^{2-}
(3) SO_3 , CO_3^{2-} (4) SO_4^{2-} , CO_3^{2-}
88. The hybridisation scheme for the central atom includes a d-orbital contribution in
- (1) I_3^- (2) PCl_3
(3) NO_3^- (4) H_2Se
89. The shape of XeO_2F_2 molecule is
- (1) trigonal bipyramidal
(2) square planar
(3) tetrahedral
(4) see-saw
90. 10 mL each of solution with pH values 1.3, 2.4, 3.6, 4.8 are mixed, then the pH of mixture will be close to :-
- (1) 1.9 (2) 2.5
(3) 3.2 (4) 4.2

SPACE FOR ROUGH WORK

91. Find out the correct match.
- (1) Apoenzyme = Co-factor + Co-enzyme
 - (2) Conjugated enzyme = Simple enzyme + apoenzyme
 - (3) Prosthetic group = Co-factor - Co-enzyme
 - (4) Holoenzyme = Apoenzyme + Co-factor
92. Which of the following statements regarding enzyme inhibition are incorrect?
- A. Competitive inhibition occurs when a substrate competes with enzyme for binding to inhibitor protein.
 - B. Competitive inhibition occurs when the substrate and the inhibitor compete for active site on enzyme.
 - C. Non-competitive inhibitor, of an enzyme can be overcome by adding large amount of substrate.
 - D. Non-competitive inhibitor often bind to the enzyme irreversibly, so it decreases K_m constant
- (1) A, B and C
 - (2) A, C and D
 - (3) B, C and D
 - (4) A, B and D
93. Zinc is the cofactor of enzyme
- (1) Hexokinase
 - (2) Amylase
 - (3) Carboxypeptidase
 - (4) Urease
94. The substrate concentration at which the chemical reaction catalyzed by an enzyme attains half of its maximum velocity is termed as
- (1) $1/2 V_{max}$
 - (2) K_m
 - (3) $1/2 K_m$
 - (4) V_{max}
95. The effect of Glucose-6-phosphate on enzyme hexokinase is an example of
- (1) Allosteric modulation
 - (2) Feed-back inhibition
 - (3) End product inhibition
 - (4) All of these
96. Which of the following is the best evidence for the lock and key theory of enzyme action?

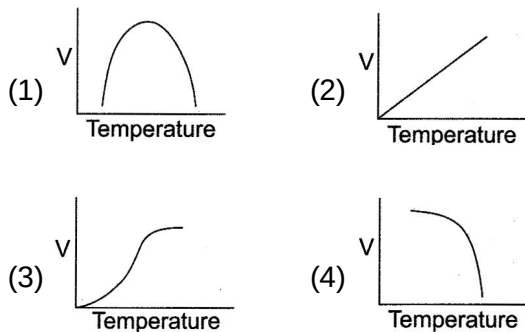
- (1) Compounds similar in structure to the substrate inhibit the reaction
- (2) Enzymes determine the direction of a reaction
- (3) Enzymes are found in living organisms and speed up certain reactions
- (4) All isolated enzymes have been identified as proteins

97. The figure given below shows the conversion of a substrate into product by an enzyme. Which of the following is the correct option for labellings A, B, C and D?



	A	B	C	D
(1)	Potential energy	Activation energy with enzyme	Activation energy without enzyme	Transition state
(2)	Transition state	Potential energy	Activation energy without enzyme	Activation energy with enzyme
(3)	Transition state	Potential energy	Activation energy with enzyme	Activation energy without enzyme
(4)	Potential energy	Transition State	Activation energy with enzyme	Activation energy without enzyme

98. Which one of the following graphs depicts the change in enzymes activity with variation in temperature?



99. Enzymes, vitamins and minerals are common in

- (1) Being synthesised in the body of organisms
- (2) Regulating metabolism
- (3) Being biocatalyst
- (4) Being proteinaceous in nature

100. Which of the following enzyme does not belong to the category hydrolase?

- (1) Dehydrogenase
- (2) Amylase
- (3) Sucrase
- (4) Protease

101. How many of the following statements are incorrect :-

- (A) Floral formula shows cohesion and adhesion within parts of whorls and between whorls.
- (B) Solanaceae is widely distributed in tropics, subtropics, and even temperature zones.
- (C) Many plants belonging to family liliaceae are good ornamentals, source of medicine, vegetables and colchicine
- (D) Generally dicotyledonous plants have tap roots while monocotyledonous plants have fibrous roots.
- (E) In banana, pineapple, Pistia and chrysanthemum, the lateral branches originate from the basal and underground portion of the main stem, grow horizontally beneath the soil, and then come out obliquely upward given rise to leafy shoots.
- (F) Photosynthesis is restricted to tendrils in opuntia and peas.

- (1) 0 (2) 1 (3) 2 (4) 3

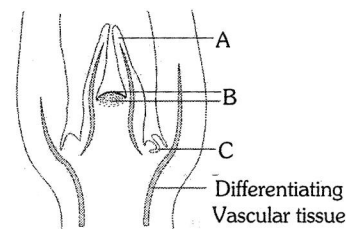


floral formula given above, is concerned with how many of the following :-

- (A) Replum
- (B) Marginal Placentation
- (C) Tetradynamous condition
- (D) Diadelphous condition
- (E) Silicula
- (F) Cyathium
- (G) Syncarpous
- (H) Apocarpous
- (I) Superior ovary
- (J) Hypogynous flower

- (1) 5 (2) 6
(3) 7 (4) 8

103. Identify the following points A, B and C in the given figure :-



- (1) A - Root hair primordium, B - Root apical meristem, C - Terminal bud
- (2) A - Root hair primordium, B - Root apical meristem, C - Axillary bud
- (3) A - Leaf primordium, B - Shoot apical meristem, C - Apical bud
- (4) A - Leaf primordium, B - Shoot apical meristem, C - Axillary bud

104. Grafting is not possible in monocots because they

- (1) Have scattered vascular bundles
- (2) Lack cambium
- (3) Are herbaceous
- (4) Have parallel venation

105. The composition of stele is

- (1) Endodermis, pericycle
- (2) Pericycle, pith
- (3) Pith, vascular bundle
- (4) Endodermis, pericycle, pith

106. Assertion : Lateral meristems are cylindrical meristem.

Reason : This meristems occur in mature region of root and shoots of many plants.

(1) If both Assertion & Reason are true & the reason is a correct explanation of the assertion.

(2) If both Assertion & Reason are true but reason is not a correct explanation of the arrertion.

(3) If Assertion is true but the Reason is false

(4) If both Assertion & Reason are false

107. Bulliform cells

(I) Large, empty, colourless cells

(II) Are certain abaxial epidermal cells

(III) Found in grasses

(IV) During water stress, they make the leaf curl inward to minimize water loss

Select correct option.

(1) (I), (II), (IV) are correct

(2) Only (I) is incorrect

(3) All are correct except (II)

(4) (I), (II), (II) and (IV) all are correct

108. Select the incorrect statement w.r.t. phloem.

(1) Complex permanent tissue which transport food materials

(2) In gymnosperm phloem lack sieve tubes and companion cells

(3) Protophloem has bigger sieve tubes

(4) Phloem fibres are absent in primary phloem

109. Fill in the blanks correctly.

(A) Vascular cambium cut off cells towards pith that mature into _____

(B) Primary and secondary phloem gradually get crushed due to continued formation and accumulation of _____

(C) The vascular cambium is generally more active on the _____ side

(D) Cells of cortex dedifferentiate and form extra stellar cambium called _____

(1) A - Secondary xylem B - Secondary xylem C - Inner D - Cork cambium

(2) A - Secondary phloem B - Secondary xylem C - Outer D - Phellogen

(3) A - Secondary xylem B - Phellem C - Inner side D - Phellogen

(4) A - Secondary xylem B - Phelloderm C - Inner side D - Cork cambium

110. If a cross section of tree of temperate region passing through base of a tree shows 100 rings of light colour and 100 rings of dark colour, then the age of the tree is

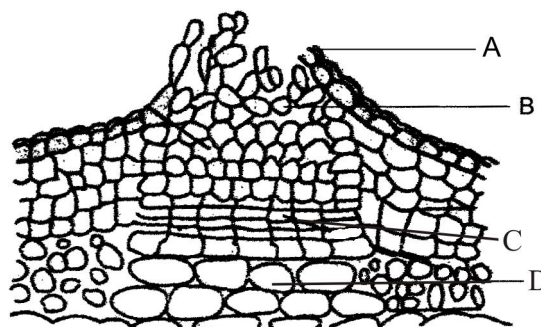
(1) 50 years

(2) 100 years

(3) 99 years

(4) Cannot be calculated

111. In the given diagram, identify A, B, C and D.



	A	B	C	D
(1)	Epidermis	Complementary cells	Phellogen	Phelloderm
(2)	Epidermis	Phellem	Phellogen	Phelloderm
(3)	Phellogen	Periderm	Phellogen	Phellem
(4)	Epidermis	Cork	Cork cambium	Secondary cortex

112. Identify the incorrectly matched pair w.r.t. anatomy of organs.

Organ	Feature
(1) Dicot stem	Eustelic condition
(2) Monocot stem	Protoxylem cavities within the vascular bundle
(3) Monocot root	Diarch to hexarch arrangement of xylem
(4) Dicot leaf	Dorsiventral leaf

113. Oil ducts in stem of sunflower are

- (1) Lysigenous
- (2) Schizogenous
- (3) Schizolysigenous
- (4) Both (2) and (3)

114. Monocot stem is similar to dicot stem in presence of

- (1) Phloem parenchyma
- (2) Semilunar patches of sclerenchyma above phloem
- (3) Conjoint and collateral vascular bundle
- (4) Sclerenchymatous bundle sheath

115. Assertion: Both apical meristem and intercalary meristems are primary meristems.

Reason: They appear early in life of a plant and contribute to the formation of the primary plant body.

- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

116. Assertion : The companion cells help in maintaining the pressure gradient in sieve tube.

Reason : The companion cells are specialised parenchymatous cells, which are closely associated with sieve tube elements.

- (1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of the assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

117. Lateral meristems are

- (1) Phellogen and procambium
- (2) Fascicular cambium and cork cambium
- (3) Fascicular cambium and procambium
- (4) Procambium and dermatogens

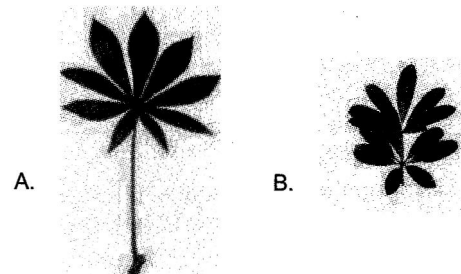
118. Which one among the following is correct?

- (1) Tracheids are unicellular with wide lumen
- (2) Tracheids are multicellular with narrow lumen
- (3) Vessels are multicellular with wide lumen
- (4) Vessels are unicellular with narrow lumen

119. Ground tissue consists of

- (1) All tissues internal to endodermis
- (2) Epidermis and cortex
- (3) All tissues except epidermis and vascular bundles
- (4) All tissue external to endodermis

120. Examine the figure (A) and (B) and choose correct option :-



- (1) A – Palmately compound leaf present in *Nerium*
- (2) B – Whorled phyllotaxy present in *Alstonia*
- (3) A – Whorled phyllotaxy present in silk cotton
- (4) B – Palmately compound leaf present in *Alstonia*

121. Tendril of pea and tendril of gourds :-

- (1) Are structurally and functionally similar
- (2) Ontogenically similar
- (3) Homologous organ
- (4) Are similar in function

122. Type of inflorescence where the main axis is limited in growth :-

- (1) Called cymose
- (2) Shows basipetal arrangement of flower
- (3) Bear terminal flower
- (4) All of these

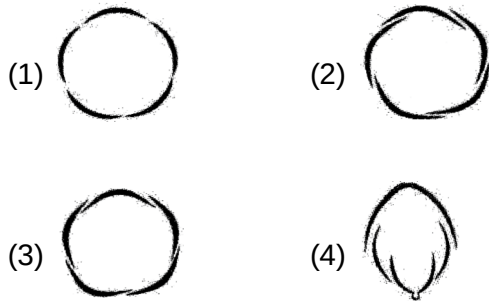
123. There may be a variation in the length of filament with in flower, as in :-

- (1) Chinrose and cotton
- (2) Mustard and *Salvia*
- (3) *Salvia* and pea
- (4) Makoi and brinjal

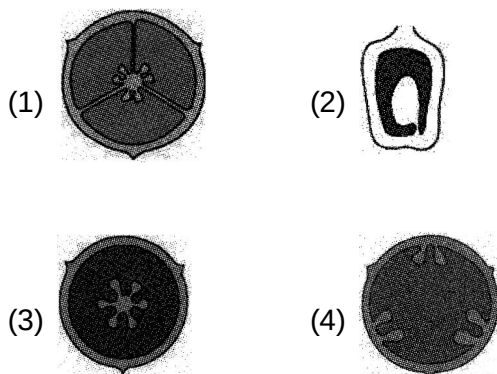
124. Both cyathium and hypanthodium inflorescence possess

- (1) Bisexual sessile flowers
- (2) Unisexual flowers
- (3) Polygamous flowers
- (4) Neuter flowers

125. Aestivation of petals in the gulmohar is correctly shown in :-



126. Which of the following diagram represent the placentation in which placenta develops at the base of ovary?



127. Spikelet is commonly seen in :-

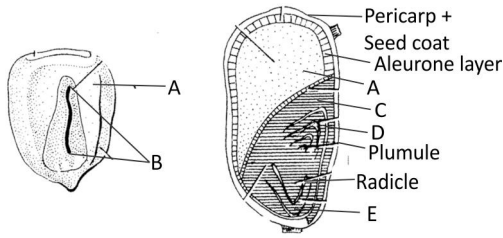
- (1) Radish
- (2) Mustard
- (3) Cauliflower
- (4) Wheat

128. Match the columns I and II, and choose the correct combination from the options given.

Column-I		Column-II	
(a)		1.	Brassicaceae
(b)		2.	Fabaceae
(c)		3.	Solanaceae
(d)		4.	Liliaceae

- (1) a-3, b-2, c-1, d-4
- (2) a-1, b-3, c-4, d-2
- (3) a-2, b-4, c-1, d-3
- (4) a-2, b-4, c-3, d-1

129. The following diagram is the typical structure of monocotyledonous seeds. Identify all the five parts A, B, C, D and E



- (1) A-Embryo, B-Endosperm, C-Scutellum, D-Coleorhiza, E-Coleoptile
- (2) A-Endosperm, B-Embryo, C-Scutellum, D-Coleoptile, E-Coleorhiza
- (3) A-Embryo, B-Endosperm, C-Scutellum, D-Coleoptile, E-Coleorhiza
- (4) A-Endosperm, B-Embryo, C-Scutellum, D-Coleorhiza, E-Coleoptile

130. An aspect of flower shown in floral formula but not in floral diagram is

- (1) Position of ovary
- (2) Floral symmetry
- (3) Aestivation
- (4) Cohesion of floral parts

131. Match the columns I, II and III and choose the correct combination from the options given.

Column I	Column II	Column III
a. $\text{Br} \oplus \text{P}_{(3+3)} \text{A}_{(3+3)} \text{G}_{(3)}$	1. Solanaceae	K. Mustard
b. $\% \oplus \text{K}_{(5)} \text{C}_{1+2+2} \text{A}_{(9)+1} \text{G}_1$	2. Liliaceae	L. <i>Indigofera</i>
c. $\oplus \text{K}_{(5)} \text{C}_{(5)} \text{A}_5 \text{G}_{(2)}$	3. Brassicaceae	M. <i>Petunia</i>
d. $\oplus \text{K}_{2+2} \text{C}_4 \text{A}_{2+4} \text{G}_{(2)}$	4. Fabaceae	N. <i>Tulip</i>

- (1) a-2-M, b-1-L, c-3-K, d-4-N
- (2) a-2-M, b-4-N, c-1-L, d-3-K
- (3) a-2-N, b-4-L, c-1-M, d-3-K
- (4) a-2-N, b-4-M, c-3-K, d-1-L

132. In china rose the flowers are

- (1) Zygomorphic, epigynous with twisted aestivation
- (2) Actinomorphic, hypogynous with twisted aestivation
- (3) Actinomorphic, epigynous with valvate aestivation
- (4) Zygomorphic, hypogynous with imbricate aestivation

133. Glumes are modified

- (1) Petals
- (2) Bracts
- (3) Gynoecium
- (4) Androecium

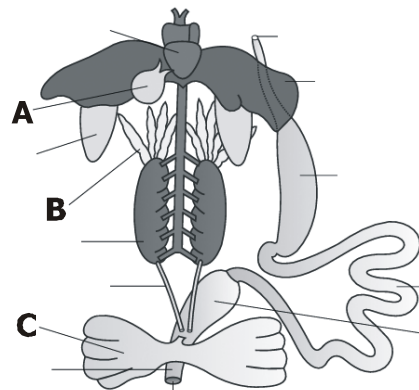
134. When placenta forms a ridge along the ventral suture of the ovary and the ovules are borne on this ridge forming two rows, the placentation is termed as

- (1) Axile
- (2) Parietal
- (3) Marginal
- (4) Basal

135. A flower is zygomorphic when

- (1) Any transverse section divides it into two equal halves
- (2) Only one transverse section divides it into two equal halves
- (3) Every vertical section passing through its centre divides it into two equal halves
- (4) Only one vertical section passing through its centre divides it into two equal halves

136. Frog belonging to class of phylum chordata are animals and show hibernation in and aestivation in :
- Amphibia, Poikilothermal, winter, summer
 - Amphibia, Poikilothermal, summer, winter
 - Poikilothermal, Amphibia, winter, summer
 - Poikilothermal, Amphibia, summer, winter
137. One of the main function of frog skin is :
- Diffusion of respiratory gases
 - Absorption of ultra violet rays to produce vit-D
 - Storage of excess food in the form of subcutaneous fat
 - Excretion of nitrogenous waste in the form of uric acid
138. What does not found in skin of frog :
- Scale
 - Tympanic membrane
 - Poison gland
 - Mucous gland
139. The structure present in man but absent in frog is :
- Pancreas
 - Salivary gland
 - Thyroid gland
 - Adrenal gland
140. In frog cutaneous respiration occurs :
- Always
 - Only on land
 - Only in water
 - Only during aestivation or hibernation
141. Frog heart consist of :
- Two auricles, one sinus venosus, one conus arteriosus and one ventricle
 - One auricle and two ventricles
 - Two auricles and two ventricles
 - One sinus venosus, one auricle & 2 ventricles
142. In frog, the excretory system consists of :
- Kidney, Ureters and Urinary bladder only
 - Kidney and Urinary bladder only
 - Kidney, Ureters, Urinary bladder and cloaca only
 - Kidney and cloaca only
143. Which of the following is false about the nervous system of frog :
- Fore brain includes paired olfactory lobes, paired cerebral hemispheres and paired diencephalon
 - Mid brain comprises paired optic lobes and paired crura cerebri
 - Hind brain comprises cerebellum & medulla oblongata
 - There are 9 pair of spinal nerves
144. Bidder canal is meant for passage of :
- Ova
 - Urine
 - Sperm
 - Both (2) and (3)
145. In male frog, cloaca is used to pass out :
- Faecal matter only
 - Urine only
 - Sperm only
 - Faeces, Urine and sperm
146. Compare to that of humans, the erythrocytes in frog are :
- without nucleus but with Hb
 - Nucleated and with Hb
 - Nucleated and without Hb
 - Very much smaller and fewer
147. Identify A, B & C respectively :



- Gall bladder, Fat bodies, Urinary bladder
- Lung, Fat bodies, Urinary bladder
- Gall bladder, Fat bodies, Cloaca
- Gall bladder, Liver, Urinary bladder

148. In male frog 10-12 vasa efferentia enter the kidney and open into :

- (1) Urinogenital duct
- (2) Seminal vesicle
- (3) Spermatophore
- (4) Bidder's canal

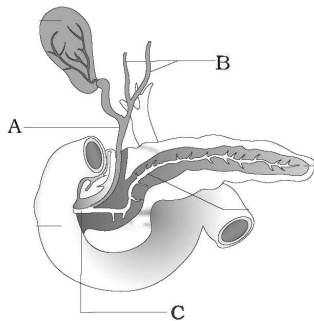
149. Which of the following is false in reference to frog ?

- (1) Fertilization is external
- (2) Development involves larval stage called tadpole
- (3) Tadpole undergo metamorphosis to form the adult
- (4) Development in frog is direct

150. Frog are beneficial for man kind because they :

- (1) eat insects and protect the crop
- (2) maintain ecological balance
- (3) the muscular leg of frog are used as food
- (4) all of these

151. Label the given diagram with A, B & C :



- (1) A = Hepato-pancreatic duct, B = Right & left hepatic duct, C = Common bile duct
- (2) A = Right & left hepatic duct, B = Hepato-pancreatic duct, C = Common bile duct
- (3) A = Common bile duct, B = Right & left hepatic duct, C = Hepato-pancreatic duct
- (4) A = Right & left hepatic duct, B = Common bile duct, C = Gall bladder

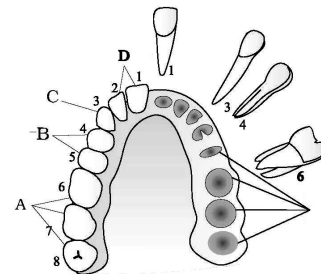
152. Which of following is absent in saliva ?

- (1) Electrolytes (Na^+ , K^+ , Cl^- , HCO_3^-)
- (2) Protease
- (3) α - salivary amylase
- (4) Lysozymes

153. The enzymes which are optimally active in acidic medium are :

- (1) Salivary amylase and Pepsin
- (2) Brush border enzymes
- (3) Trypsin
- (4) All of these

154. Identify A, B, C & D in the given diagram of human teeth



- (1) A = Incisor, B = Molar, C = Canine, D = Premolar
- (2) A = Molar, B = Premolar, C = Canine, D = Incisor
- (3) A = Premolar, B = Molar, C = Incisor, D = Canine
- (4) A = Canine, B = Incisor, C = Premolar, D = Molar

155. Which of the following component of succus entericus is not secreted from brush border cells ?

- (1) Mucus
- (2) Nucleosidase & Maltase
- (3) Sucrase & lactase
- (4) Dipeptidases & lipases

156. Select an incorrect statement about the lipid absorption

- (1) Micelles are formed in cavity of small intestine.
- (2) Chylomicrons are synthesized in intestinal cells
- (3) Bile salts help in micelle formation
- (4) Chylomicrons are absorbed mainly in blood.

157. Vomiting centre to control the vomiting reflex is present in :
 (1) Hypothalamus
 (2) Medulla
 (3) Mid-brain
 (4) Cerebellum
158. Castale's intrinsic factor helps in absorption of and secreted by cells of gastric glands :
 (1) Cyanocobalamine — Chief cells
 (2) Vitamin B₁₂ — Oxyntic cells
 (3) Vitamin B₁₂ — Parietal cells
 (4) Any two of these
159. Which of following is not a common symptom in both PEM-diseases, marasmus and Kwashiorkar ?
 (1) Wasting of muscles
 (2) Protein deficiency
 (3) Failure of growth and brain development
 (4) Extensive oedema and swelling of body parts
160. Which of following does not occur during the inspiration
 (1) Contraction of phrenic muscle
 (2) Increase in thoracic cavity
 (3) Contraction of internal intercostal muscle
 (4) Contraction of external intercostal muscle.
161. The maximum volume of air a person can breathe in after a forced expiration is :
 (1) Vital capacity (ERV + TV + IRV)
 (2) FRC (ERV + RV)
 (3) EC (TV + ERV)
 (4) IRV (2500 - 3000 ml)
162. Select an incorrect statement :
 (1) pO₂ of oxygenated blood is 95 mm Hg.
 (2) CO₂ is 10 times more soluble than O₂.
 (3) 70% of CO₂ is transported as bicarbonates
 (4) 97% of O₂ is transported as oxyhaemoglobin by RBCs.
163. Select a condition which causes the shifting of oxyhaemoglobin dissociation curve to the left :
 (1) High pCO₂ value
 (2) High temperature
 (3) Lesser H⁺ concentration
 (4) Decreased pO₂ value
164. What amount of oxygen is delivered to the tissue by 100 ml blood under normal physiological condition ?
 (1) 5 ml (2) 10 ml
 (3) 15 ml (4) 20 ml
165. Respiratory rhythm centre of medulla is primarily responsible for :
 (1) Regulation of breathing
 (2) Increased breathing
 (3) Decreased breathing
 (4) None of these
166. Which of following have insignificant role in regulation of respiratory rhythm ?
 (1) CO₂
 (2) H⁺ concentration
 (3) O₂
 (4) Acidity
167. Which is the occupational pulmonary disease ?
 (1) Pneumonia
 (2) Emphysema
 (3) Silicosis
 (4) Lung cancer
168. Release of pancreatic juice is stimulated by :
 (1) enterokinase
 (2) enterogastrone
 (3) trypsinogen
 (4) secretin
169. Emulsification of fat is carried out by :
 (1) bile pigments
 (2) bile salts
 (3) HCl
 (4) pancreatic juice

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| <p>170. In man the zymogen or chief cells are mainly found in:</p> <ol style="list-style-type: none"> cardiac part of stomach pyloric part of stomach duodenum fundic part of stomach <p>171. Pancreatic juice and hormones of pancreas are produced by :</p> <ol style="list-style-type: none"> same cells same cells at different times statement is wrong different cells <p>172. Where is protein digestion accomplished?</p> <ol style="list-style-type: none"> stomach ileum rectum duodenum <p>173. Brunner's glands occur in :</p> <ol style="list-style-type: none"> submucosa of duodenum submucosa of stomach mucosa of oesophagus mucosa of ileum <p>174. Most of the fat digestion occurs in :</p> <ol style="list-style-type: none"> rectum stomach duodenum large intestine <p>175. Which of the following is the function of enterogastrone?</p> <ol style="list-style-type: none"> it inhibits the secretion of gastric juice it stimulates the secretion of digestive juices in the stomach it stimulates the flow of pancreatic juice it regulates the flow of bile | <p>176. What is common among amylase, rennin and trypsin?</p> <ol style="list-style-type: none"> these are produced in stomach these act at a pH lower than 7 these all are proteins these all are proteolytic enzymes <p>177. If pancreas is removed, the compound which remain undigested is :</p> <ol style="list-style-type: none"> proteins carbohydrates fats all these <p>178. In vertebrates, lacteals are found in :</p> <ol style="list-style-type: none"> oesophagus ear ileum ischium <p>179. Which of the following carries glucose from digestive tract to liver?</p> <ol style="list-style-type: none"> pulmonary vein hepatic artery hepatic portal vein none of these <p>180. Inhibition of gastric & Stimulation of gastric pancreatic & bile secretions are controlled by hormones :</p> <ol style="list-style-type: none"> Gastrin , Secretin , Enterokinins & Cholecystokinin Enterogasterone, gastrin, Pancreozymin & Cholecystokinin Gastrin, enterogasterone, cholecystokinin & Pancreozymin Secretion, enterogasterone, gastrin & enterokinins |
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